CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
OPERATING SYSTEMS (CT104H)
FINAL LAB

Declaration of own work

I, NGUYEN MINH NHUT B2205896 , certify that this assignment is my own work, is not copied from any other person's work

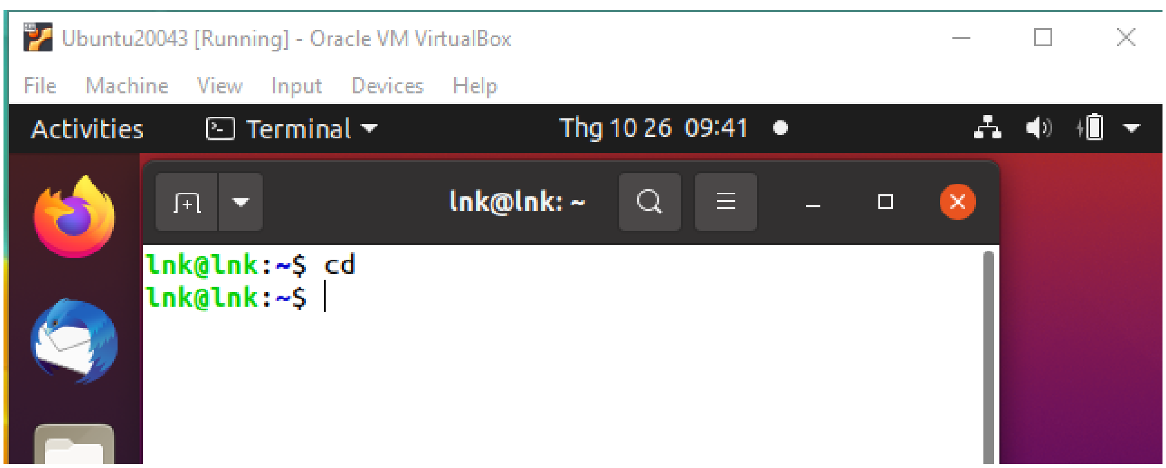
Submission: Students submit these following files to the class on Elearning system (where *StudentName* is the student's name, and *ID* is the student's ID).

* **01 file** named *StudentName_ID_CT104H_FinalLab_report.pdf*
* **01 file** named *StudentName_ID_CT104H_FinalLab_code.rar* (containing all codes)

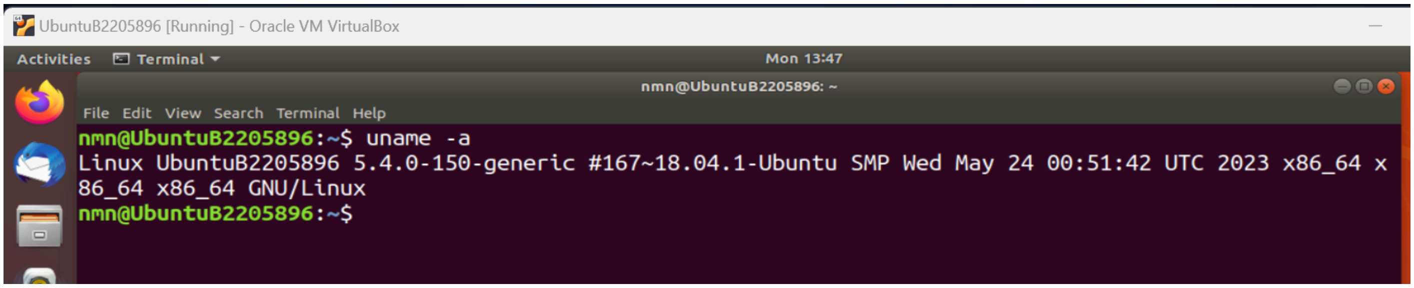
Instructions on how to present in the report file

- *Students creates an Ubuntu Virtual machine named UbuntuID (ID is the student's ID).* For example, we have a virtual machine named Ubuntu20043, such that the student *ID* is 20043.
- For each question, students MUST provide the commands/scripts AND screenshots of the commands used and/or the content of files/scripts, CLEARLY. Note*: the screenshots need to show the name of the Ubuntu Virtual machine.*
- *Do not change the directory*
- *Replace **YourName** with your full name*

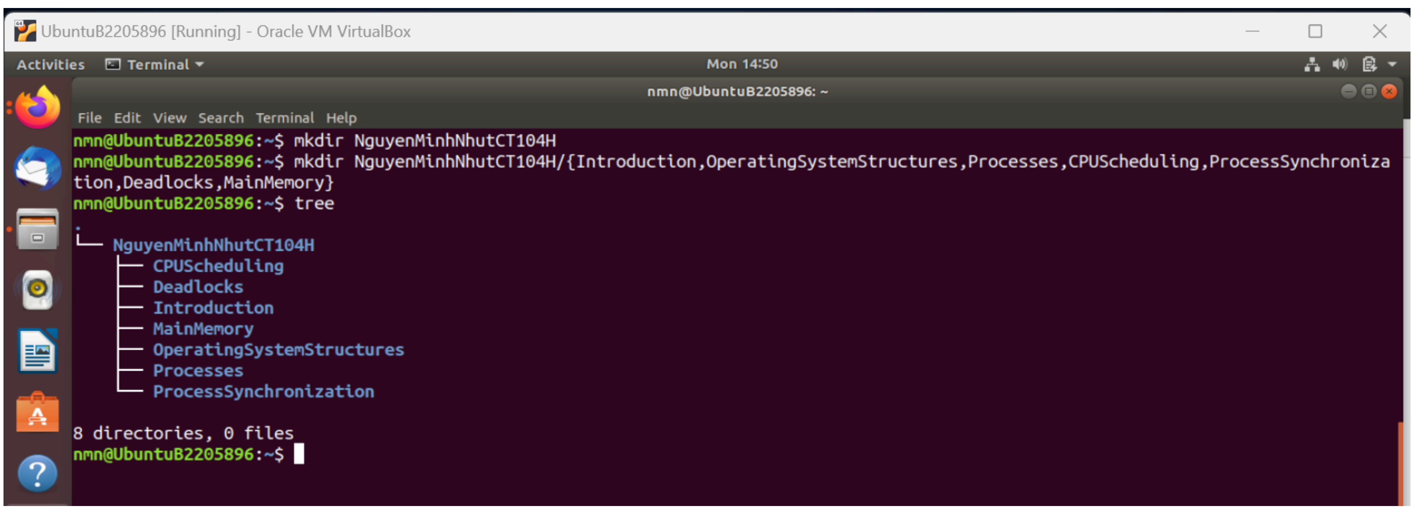
Question 0: Navigate to your home directory
Answer: $cd



<u>Question 1</u> (1.0 point - Use 1 command): List all information about your kernel system.
Answer:
$ uname -a

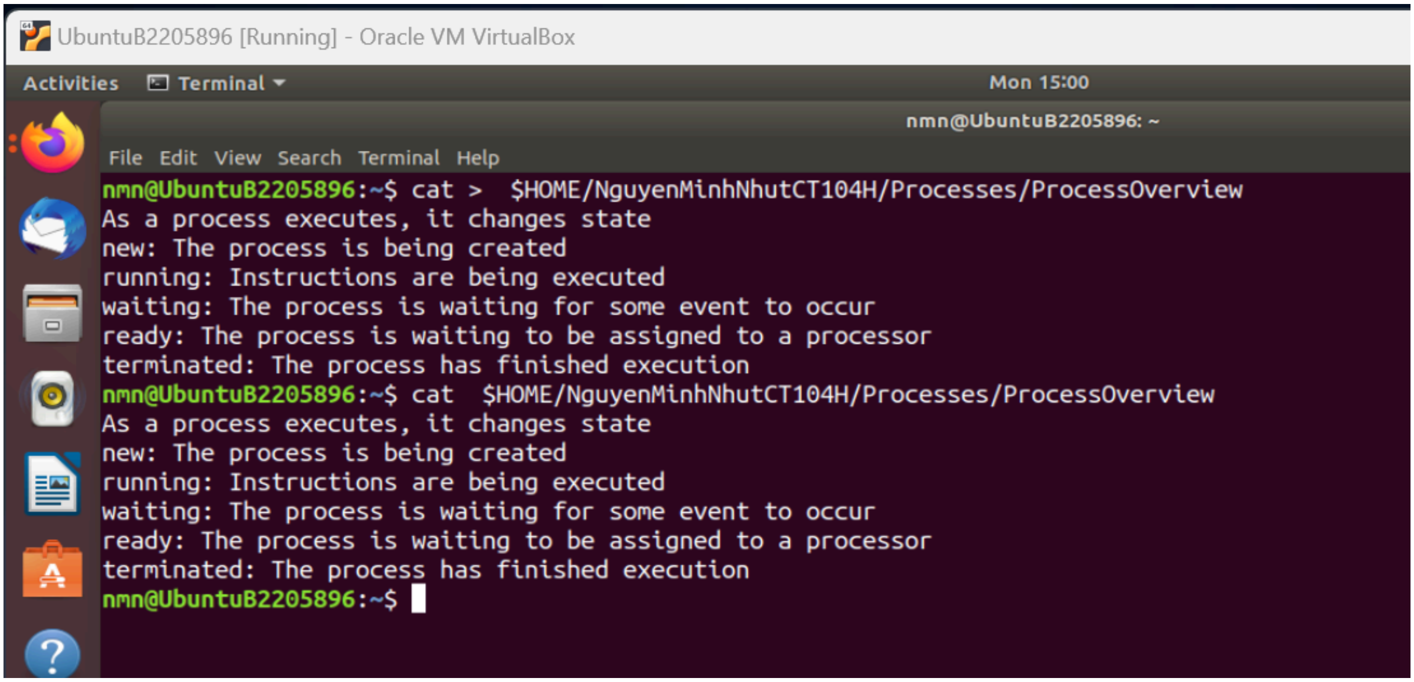


<u>Question 2</u> (1.0 point - Use 2 commands to solve this question): Create a folder named ***YourName*CT104H** that contains a list of folders, each of which is named by the title of the chapter you have studied in the CT104H class (e.g., Introduction, OperatingSystemStructures, Processes…)
Answer:
$ mkdir NguyenMinhNhutCT104H
$ mkdir NguyenMinhNhutCT104H/{Introduction,OperatingSystemStructures,Processes,CPUScheduling, ProcessSynchronization,Deadlocks,MainMemory}



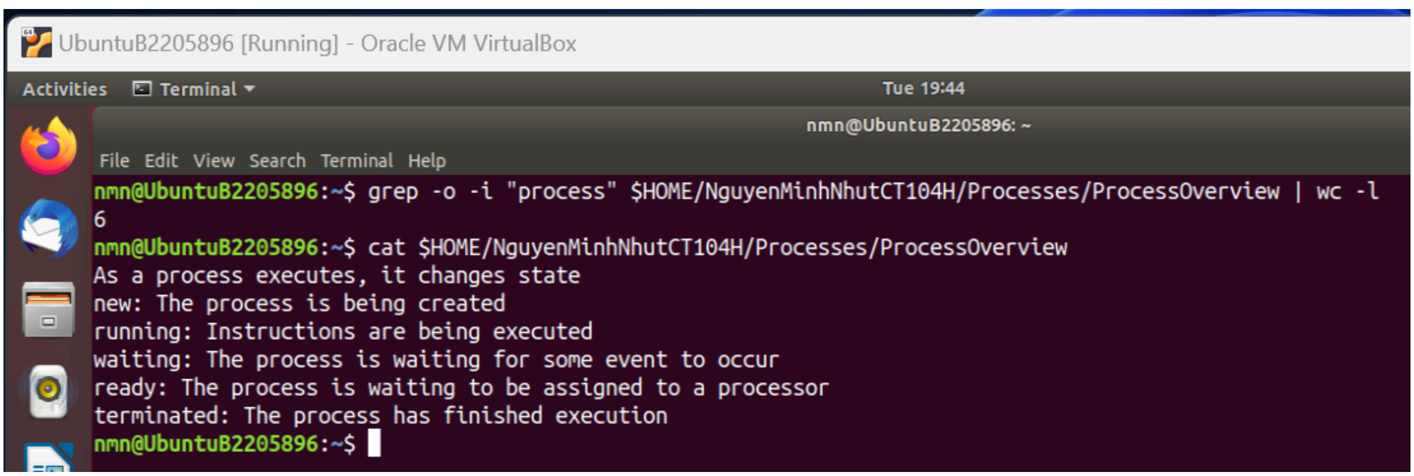
<u>Question 3</u> (1.0 point - Use 1 command to solve this question, and do not use any text editors): Inside the folder **Processes**, create a file named **ProcessOverview** such that the content of the file is the states of a process, and a brief explanation of the state. Note that, each state is presented in a single line.
Answer:
$ cat > $HOME/NguyenMinhNhutCT104H/Processes/ProcessOverview
As a process executes, it changes state
new: The process is being created
running: Instructions are being executed
waiting: The process is waiting for some event to occur
ready: The process is waiting to be assigned to a processor
terminated: The process has finished execution

$ cat $HOME/NguyenMinhNhutCT104H/Processes/ProcessOverview



<u>Question 4</u> (1.0 point - Use 1 command to solve this question): Print the occurrence count of the string "process" (regardless of uppercase or lowercase) in the file **ProcessOveriew**.

Answer:
$ grep -o -i "process" $HOME/NguyenMinhNhutCT104H/Processes/ProcessOverview | wc -l



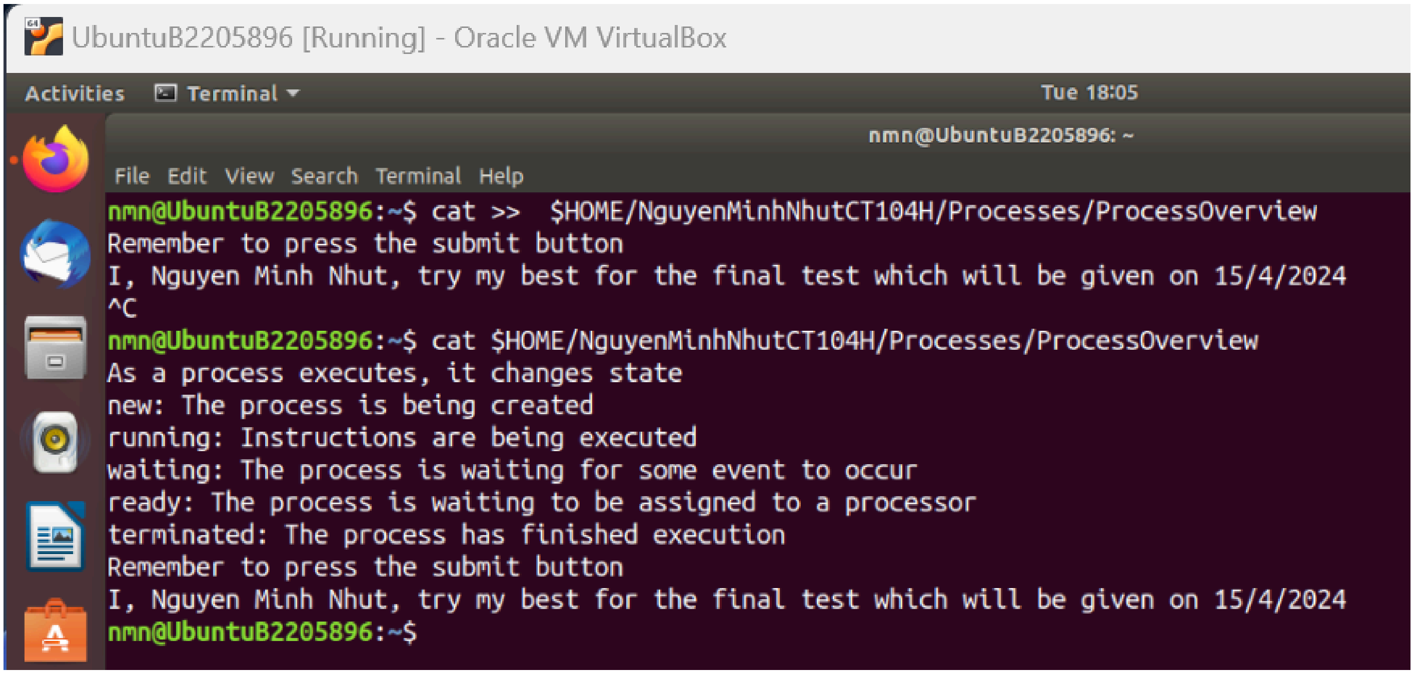
<u>Question 5</u> (1.0 point - Use 1 command to solve this question, and do not use any text editors): Add the following 2 lines to the end of the file **ProcessOverview**:

Remember to press the submit button I, *YourName,* try my best for the final test which will be given on *theDateOfYourOSFinalTest*

(Note: Please check the date of your final exam on the Elearning system.)

Answer:
$ cat >> $HOME/NguyenMinhNhutCT104H/Processes/ProcessOverview
Remember to press the submit button
I, Nguyen Minh Nhut, try my best for the final test which will be given on 15/4/2024
(Ctrl + c to save the change)

$ cat $HOME/NguyenMinhNhutCT104H/Processes/ProcessOverview

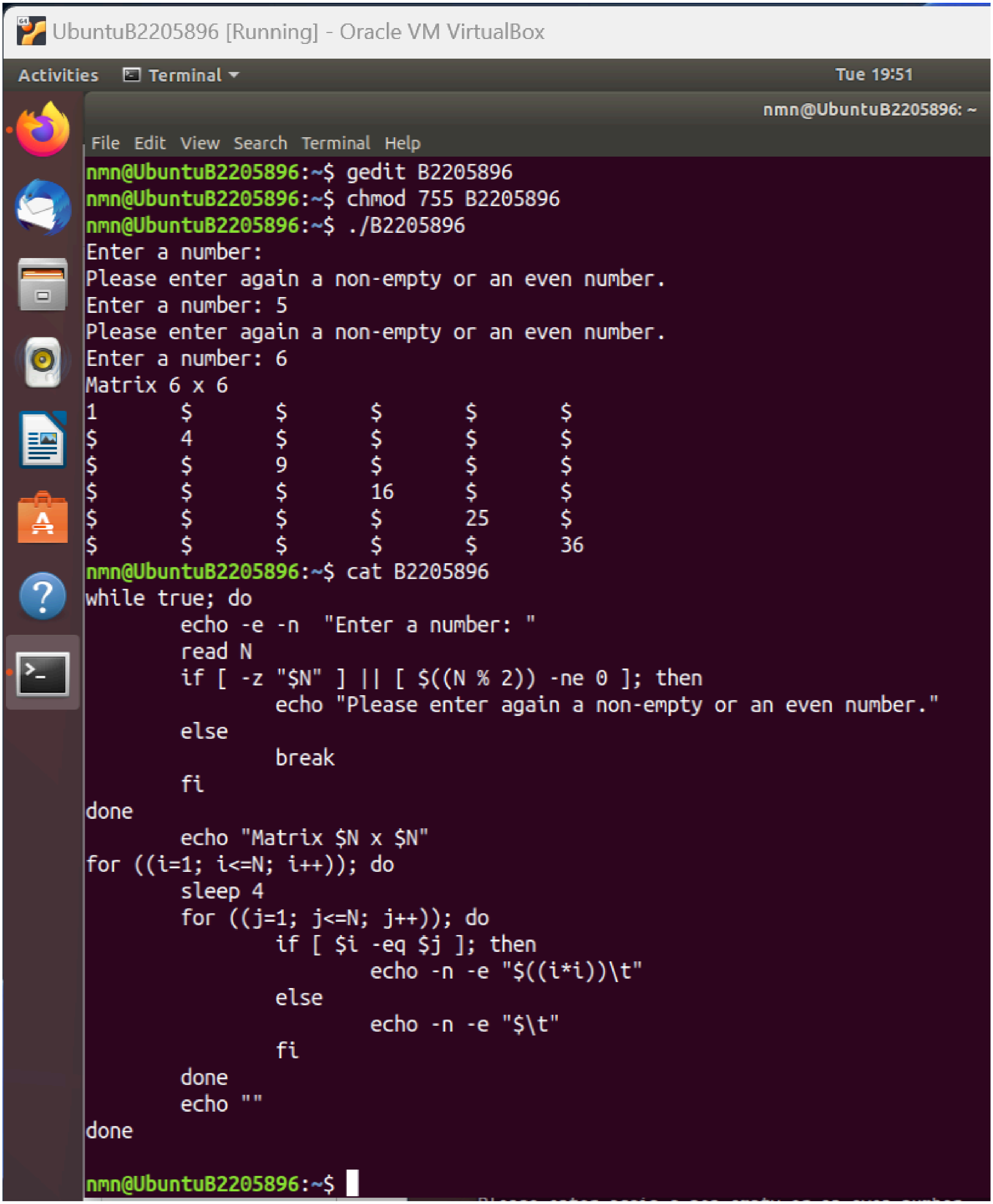


__**Question 6**__ (5.0 points): Write a shell script, named *YourStudentID,* that allows a user to input a number *N*, and then performs the following check:

- If *N* is empty or an odd number, then ask the user to reinput *N*
- Otherwise, print a matrix *N*x*N* such that all elements are $, except the elements on the diagonal line, each of which has a value of ***[i][i]=i²***, In addition, each line will appear after 4 seconds.

1	$	$	$	$	$
$	4	$	$	$	$
$	$	9	$	$	$
$	$	$	16	$	$
$	$	$	$	25	$
$	$	$	$	$	36

Answer:
$ gedit B2205896
$ chmod 755 B2205896
$ ./B2205896
(input 5, 6 to see the result)
$ cat B2205896 (show the script)




```
nmn@UbuntuB2205896:~$ gedit B2205896
nmn@UbuntuB2205896:~$ chmod 755 B2205896
nmn@UbuntuB2205896:~$ ./B2205896
Enter a number:
Please enter again a non-empty or an even number.
Enter a number: 5
Please enter again a non-empty or an even number.
Enter a number: 6
Matrix 6 x 6
1       $       $       $       $       $
$       4       $       $       $       $
$       $       9       $       $       $
$       $       $       16      $       $
$       $       $       $       25      $
$       $       $       $       $       36
nmn@UbuntuB2205896:~$ cat B2205896
while true; do
        echo -e -n  "Enter a number: "
        read N
        if [ -z "$N" ] || [ $((N % 2)) -ne 0 ]; then
                echo "Please enter again a non-empty or an even number."
        else
                break
        fi
done
        echo "Matrix $N x $N"
for ((i=1; i<=N; i++)); do
        sleep 4
        for ((j=1; j<=N; j++)); do
                if [ $i -eq $j ]; then
                        echo -n -e "$((i*i))\t"
                else
                        echo -n -e "$\t"
                fi
        done
        echo ""
done

nmn@UbuntuB2205896:~$
```

---------GOOD LUCK---------